Yucheng (Cecilia) Zhang

737-343-4836 | ceciliayczhang@gmail.com | Portfolio | LinkedIn | GitHub

Education

Northwestern University, M.S. Computer Science — Sep. 2026 – May 2028, Evanston, IL

The University of Texas at Austin, B.S. Mathematics; CS Minor; GPA: 3.74/4.00 — Aug. 2023 – May 2026, Austin, TX

Relevant Coursework: Data Structures, Software Design, Database Systems, Artificial Intelligence, Discrete Math, Probability & Statistics

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, R
Core CS: Data structures, object-oriented programming, algorithms, backend validation, testing, REST API design
Backend/Data: Flask, Node.js, PostgreSQL, SQL Server, SQLAlchemy, Alembic, Docker, Git, Pytest, Jest, Power BI
AI/Risk Systems: OpenAI API, Gemini, Claude, Large Language Models (LLMs), context filtering, credit risk modeling, fraud analytics

Experience

Enterprise Risk Analyst Intern — May 2025 – Dec. 2025
Toyota Industries Commercial Finance | Python, SQL Server, Azure, Power BI — Dallas, TX

- Automated monthly credit-risk model refreshes with parameterized SQL and Python workflows connected to Power BI dashboards, reducing manual reporting effort by **10%**.
- Built ETL and reconciliation logic across PayNet and internal loan servicing data, engineering borrower, contract, and asset features including inquiries, LTV ratio, account tenure, and vehicle segment.
- Validated Logistic Regression and Random Forest outputs with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** and improving high-risk identification by **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups to inform scorecard segmentation.

Financial Data Development Engineer – Fraud Detection — Jul. 2024 – Aug. 2024
Caitong Securities Ltd | SQL, Oracle, Stored Procedures — Hangzhou, China

- Designed SQL ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by **5%**.
- Implemented parameterized fraud-detection logic and customer risk scoring from transactional and behavioral streams, reducing false positives by **2%** in daily monitoring workflows.
- Organized stored-procedure logic into reusable modules so risk rules and threshold updates could be maintained without rewriting the full transaction pipeline.

Software Engineering Intern – Customer Analytics — Nov. 2023 – Jun. 2024
Joblogic-X Corporation | Python, REST APIs, Git — Dallas, TX

- Engineered a Python pipeline to scrape and process XHS social-media data and join it with transactional order systems for repeatable customer analytics storage.
- Implemented modular ranking and clustering algorithms, generating reproducible customer-priority lists for marketing and engagement workflows.
- Integrated a KOC scoring module into Meet Fresh's customer-facing chatbot through REST APIs, enabling ranked recommendations inside the existing customer workflow.

Data Scientist Assistant Intern – Image Restoration — Jul. 2023 – Sep. 2023
Chinese Academy of Sciences | TensorFlow, Cloud, GFPGAN — Beijing, China

- Deployed image-restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Built a TensorFlow 2 CNN on 3.4k train/validation/test images and achieved **80% accuracy** across 5 image categories.
- Integrated GFPGAN APIs for automated face restoration and color correction, reducing processing costs by **5%** and supporting model-comparison reporting.

Projects

CodeAssist – Full-Stack AI Feedback System | GitHub — Aug. 2025 – Present
University of Texas at Austin | Faculty Project | React, Flask, PostgreSQL, Docker — Austin, TX

- Built a full-stack AI feedback system for programming courses, connecting code editing, file uploads, submissions, and AI chat feedback for debugging, complexity reasoning, and test-case correctness.
- Developed Flask REST APIs and PostgreSQL schemas for submissions, instructor settings, AI requests, usage limits, cooldown policies, backend validation, JSON serialization, and role-based access.
- Implemented context filtering and multi-provider routing across OpenAI, Gemini, Claude, and local models so only instructor-approved code, files, tests, results, and outputs are sent to providers.
- Standardized provider and assignment-level settings, including model selection, API configuration, timeout handling, and prompt behavior for repeatable AI feedback workflows.
- Built local stress-test runners with server health checks, structured JSON reports, cleanup controls, and coverage for enrollment, submission, grading, AI feedback, and CRUD workflows.

Echo – Interactive AI Voice Simulation System — Jun. 2026 – Present
University of Texas at Austin | Mentor: Prof. Bo Xie | Next.js, TypeScript, OpenAI API — Austin, TX

- Built an event-driven Next.js system that generates context-aware voice responses from user pauses, errors, skipped fields, and task switching using session state, browser TTS, and OpenAI APIs.
- Designed cooldown, interval, and event-priority logic to prevent repetitive playback while keeping generated responses tied to current task state and user interaction timing.
- Modularized prompt templates and event-trigger logic so new voice events and task types can be added without changing the core session controller.